

Mattia Sensi

Caritro postdoctoral fellow at Università degli Studi di Trento

Quantitative researcher with a PhD in mathematics and experience building and analysing mathematical models of complex dynamical systems. Background spans stochastic processes, ODE/PDE systems, numerical methods, and data-driven modelling, with implementation in MATLAB and Python applied to forecasting, inference, and system design across biological and physical domains.

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EDUCATION

Ph.D. in Mathematics, *cum laude*, Università degli Studi di Trento. November 2017 – January 2021
Thesis: “A Geometric Singular Perturbation approach to epidemic compartmental models”
Supervisor: [Prof. Andrea Pugliese](#)

M.Sc. in Mathematics, Universiteit van Amsterdam. September 2015 – June 2017
Thesis: “Homoclinic vegetation stripes in a Klausmeier-Gray-Scott model”
Supervisor: [Prof. Dr. Arjen Doelman](#)

B.Sc. in Mathematics, Università degli Studi di Padova. September 2011 – September 2014
Thesis: “Portfolio optimization for quadratic utility function with partial information”
Supervisor: [Prof. Wolfgang J. Runggaldier](#)

EMPLOYMENT

Postdoctoral researcher in Mathematics, Università degli Studi di Trento. March 2025 – present
[Caritro postdoctoral fellow](#), Principal Investigator. Working in the group of [Prof. Andrea Pugliese](#), researching the project “*Mathematical models of infectious disease spreading in multi-host and heterogeneous populations: applications avian flu*”.
Calibrated and validated epidemic models against real-world avian influenza outbreak data (H5N1 HPAI, Northern Italy 2021-2022); performed parameter estimation and sensitivity analysis on epidemiological datasets in Python.

Postdoctoral researcher in Mathematics, Politecnico di Torino. March 2023 – February 2025
Postdoctoral researcher in the group of [Prof. Andrea Tosin](#), as part of PRIN 2020 project “*Integrated Mathematical Approaches to Socio-Epidemiological Dynamics*”.
Developed and implemented stochastic and compartmental models in MATLAB.

Postdoctoral researcher in Mathematics, Inria at Université Côte d’Azur. December 2021 – February 2023
Postdoctoral researcher in the group [MathNeuro](#), led by [Prof. Mathieu Desroches](#).
Developed and implemented deterministic models of neurotransmitter release dynamics; simulated in MATLAB and AUTO-07p.

Postdoctoral researcher in Mathematics, TU Delft. March – November 2021
Postdoctoral researcher in the group [NAS](#), led by [Prof. Piet Van Mieghem](#).
Developed and implemented deterministic and stochastic models of infectious disease spreading in temporal networks of individuals.

SELECTED PUBLICATIONS (FULL LIST OF 29 PUBLICATIONS)

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6. **E. Bernardi, T. Lorenzi, M. S. and A. Tosin.** *Heterogeneously structured compartmental models of epidemiological systems: from individual-level processes to population-scale dynamics*. Studies in Applied Mathematics 155, no. 2 (2025): 155, e70091
 5. **L. Eigentler and M. S..** *Delayed loss of stability of periodic travelling waves: insights from the analysis of essential spectra*. Journal of Theoretical Biology, 2024, 595, 111945
 4. **M. S., M. Desroches and S. Rodrigues.** *Slow-fast dynamics in a neurotransmitter release model: delayed response to a time-dependent input signal*. Physica D: Nonlinear Phenomena, 2023, 455, 133887
 3. **P. Kaklamanos, C. Kuehn, N. Popovic and M. S..** *Entry-exit functions in fast-slow systems with intersecting eigenvalues*. Journal of Dynamics and Differential Equations, 2025, 37(1), pp. 559–576, 103220
 2. **T. Lorenzi, A. Pugliese, M. S. and A. Zardini.** *Evolutionary dynamics in an SI epidemic model with phenotype-structured susceptible compartment*. Journal of Mathematical Biology, 2021, 83(6-7), 72
 1. **H. Jardón-Kojakhmetov, C. Kuehn, A. Pugliese and M. S..** *A geometric analysis of the SIR, SIRS and SIRWS epidemiological models*. Nonlinear Analysis: Real World Applications, 2021, 58, 103220

GRANTS & AWARDS

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- [Progetti Giovani GNFM 2025](#) with the project “*Analisi geometrica della diffusione spaziale di epidemie su più scale temporali*” (Geometric analysis of the spatial diffusion of epidemics on multiple time scales), coordinated by Rossella Della Marca, from August 2025 to July 2026. Participant, 2000 €
 - [Caritro postdoctoral fellowship 2024](#) with the project “*Modelli matematici di malattie infettive più ospiti e popolazioni eterogenee: applicazioni all’influenza aviaria*” (Mathematical models of infectious disease spreading in multi-host and heterogeneous populations: applications to avian flu), from March 2025 to February 2027. Principal investigator, 85000 €

SOFTWARE

Mathematical modelling and simulation: MATLAB, Python (numerical methods, stochastic processes, data analysis), Wolfram Mathematica. Experience developing and validating quantitative models; scientific computing; implementation of ODE solvers and stochastic simulations.

Text editing: Microsoft office tools, L^AT_EX.

LANGUAGES

Italian (mother tongue), English (C1), French (basic).